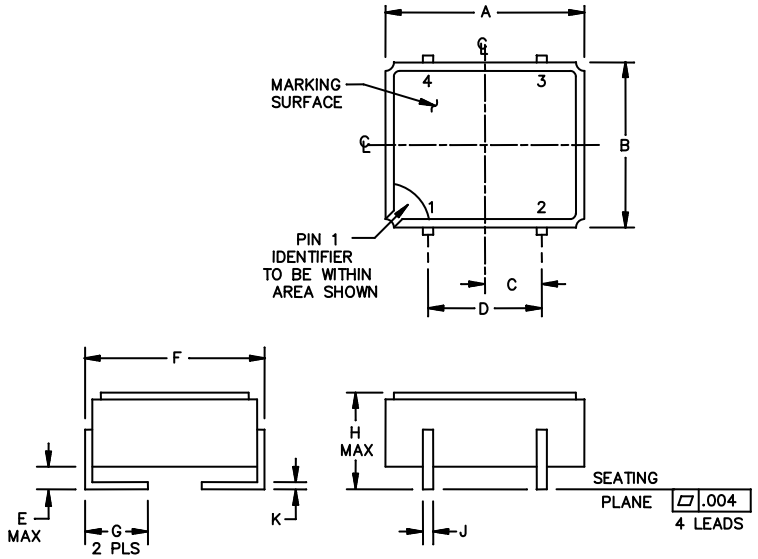
INCH-POUND
MIL-PRF-55310/38A
19 April 2010
SUPERSEDING
MIL-PRF-55310/38
23 March 2007

PERFORMANCE SPECIFICATION SHEET

# OSCILLATOR, CRYSTAL CONTROLLED, TYPE 1 (CRYSTAL OSCILLATOR (XO), 500 KHz THROUGH 150 MHz, HERMETIC SEAL, LOW VOLTAGE CMOS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55310.




| Pin number | Function |
|---|---|
| 1 | Enable/Disable |
| 2 | GND |
| 3 | Output |
| 4 | B+ |

| Ltr | Inches | | Millimeters | |
|---|---|---|---|---|
| | Min | Max | Min | Max |
| A | .345 Typ | .355 Typ | 8.80 Typ | 9.02 Typ |
| B | .285 Typ | .295 Typ | 7.24 Typ | 7.49 Typ |
| C | .086 | .114 | 2.18 | 2.90 |
| D | .195 | .205 | 4.95 | 5.20 |
| E | .050 | .060 | 1.27 | 1.52 |
| F | --- | .320 | --- | 8.13 |
| G | .105 Typ | .115 Typ | 2.67Typ | 2.92Typ |
| H | --- | .190 | --- | 4.83 |
| J | .015 | .021 | 0.39 | 0.53 |
| K | --- | .008 | --- | 0.20 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.

FIGURE 1. Interface and physical dimensions.



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REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Mounting: See figure 1.

Terminals: See figure 1.

Seal: Hermetic in accordance with MIL-PRF-55310, maximum leakage rate 5 x $10^{-8}$ atm cc/s.

Weight: 3 grams, maximum.

Oscillator: Class 2 or any class 1 or class 3 oscillator meeting all class 2 requirements and verification tests specified herein and in MIL-PRF-55310.

Calibration: Manufacturer calibrated.

Screening: In accordance with MIL-PRF-55310, product level B or S, as applicable.

Temperature:

Operating: See table I.

Storage: -62°C to +125°C.

Load test circuit: See figure 2.

Output waveform: Symmetrical square wave, 3.3 volt CMOS logic compatible (see figure 3).

Output logic voltage levels at designated CMOS (see figure 3):

Logic 1: 90 percent of $V_{DD}$, minimum.

Logic 0: 10 percent of $V_{DD}$, maximum.

Rise and fall times: (See table I) Measurements shall be taken at the 10 percent and 90 percent peak-to-peak output voltage levels, with peak-to-peak output defined as level 1-level 0 (see figure 3).

Duty cycle: See table I and figure 3.

Supply voltage: +3.3 V dc ±10 percent.

Input current: At designated supply voltage (see table I).

Enable/disable:

Output disabled (high impedance): Pin 1 input ≤ 0.3 V dc.

Output active: Pin 1 input ≥ 2.0 V dc or open.

Output frequency: Frequency as designated at time of acquisition (see table I).

Initial accuracy at reference temperature (up to 30 days after shipment): See table I.



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TABLE I. Dash numbers and operating characteristics.

| Dash number | Output frequency range | Input current (max) 2/ | Pulse characteristics | | Initial accuracy at +23°C ±1°C | Frequency-temperature tolerance (ppm) 1/ | | |
|---|---|---|---|---|---|---|---|---|
| | | | Rise and fall times (max) 3/ | Duty cycle min-max 3/ | | -55°C to +125°C | -55°C to +105°C | -20°C to +70°C |
| | | | | | | A | B | C |
| 01 | 500 kHz to 15.999 MHz | 6 mA | 4 ns | 45% to 55% | ±15 ppm | ±50 ppm | ±40 ppm | ±25 ppm |
| 04 | 500 kHz to 15.999 MHz | 6 mA | 4 ns | 45% to 55% | ±25 ppm | ±100 ppm | ±80 ppm | ±50 ppm |
| 11 | 16.000 MHz to 31.999 MHz | 10 mA | 4 ns | 40% to 60% | ±15 ppm | ±50 ppm | ±40 ppm | ±25 ppm |
| 14 | 16.000 MHz to 31.999 MHz | 10 mA | 4 ns | 40% to 60% | ±25 ppm | ±100 ppm | ±80 ppm | ±50 ppm |
| 21 | 32.000 MHz to 39.999 MHz | 15 mA | 4 ns | 40% to 60% | ±15 ppm | ±50 ppm | ±40 ppm | ±25 ppm |
| 24 | 32.000 MHz to 39.999 MHz | 15 mA | 4 ns | 40% to 60% | ±25 ppm | ±100 ppm | ±80 ppm | ±50 ppm |
| 31 | 40.000 MHz to 64.999 MHz | 20 mA | 4 ns | 40% to 60% | ±15 ppm | ±50 ppm | ±40 ppm | ±25 ppm |
| 34 | 40.000 MHz to 64.999 MHz | 20 mA | 4 ns | 40% to 60% | ±25 ppm | ±100 ppm | ±80 ppm | ±50 ppm |
| 41 | 65.000 MHz to 99.999 MHz | 30 mA | 4 ns | 40% to 60% | ±15 ppm | ±50 ppm | ±40 ppm | ±25 ppm |
| 44 | 65.000 MHz to 99.999 MHz | 30 mA | 4 ns | 40% to 60% | ±25 ppm | ±100 ppm | ±80 ppm | ±50 ppm |
| 51 | 100.00 MHz to 129.999 MHz | 40 mA | 3 ns | 40% to 60% | ±15 ppm | ±50 ppm | ±40 ppm | ±25 ppm |
| 54 | 100.00 MHz to 129.999 MHz | 40 mA | 3 ns | 40% to 60% | ±25 ppm | ±100 ppm | ±80 ppm | ±50 ppm |
| 61 | 130.00 MHz to 150.00 MHz | 45 mA | 3 ns | 40% to 60% | ±15 ppm | ±50 ppm | ±40 ppm | ±25 ppm |
| 64 | 130.00 MHz to 150.00 MHz | 45 mA | 3 ns | 40% to 60% | ±25 ppm | ±100 ppm | ±80 ppm | ±50 ppm |

1/ Temperature range A applicable for product level B oscillators only.
2/ Maximum input current for no load condition.
3/ See figure 3.

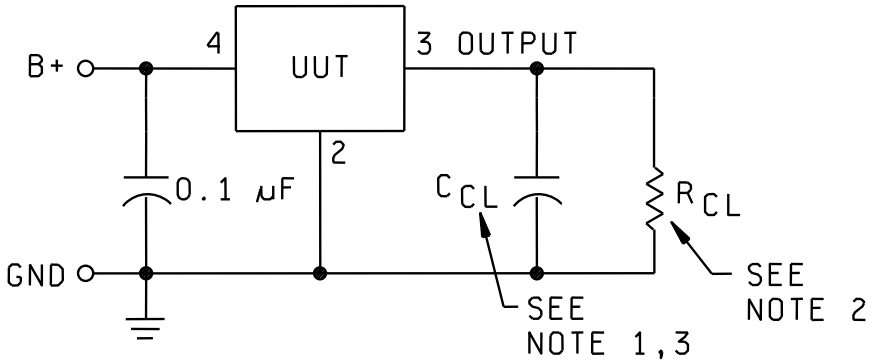
Frequency-temperature tolerance (one-half temperature cycle, referenced to frequency measured at +23°C ±1°C, immediately prior to starting of the test): See table I. Measurements taken at ten equally spaced increments over the specified operating temperature range. 1/

Frequency-voltage tolerance: ±4 ppm maximum for a ±10 percent change in supply voltage. Measurements taken at reference temperature and operating temperature range end points.

1/ For the purpose of transitioning this device to MIL-PRF-55310, 'Frequency stability versus temperature' has been renamed 'Frequency-temperature tolerance'. The verification requirements of 'initial frequency-temperature accuracy (one-half temperature cycle)' shall apply except that frequency measurements shall be referenced to the frequency measured at +23°C ±1°C ($f_{ref}$) instead of to the nominal frequency ($f_{nom}$).



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NOTES:
1. For $C_{CL}$ = 15 pF ±5 percent.
2. For $R_{CL}$ = 10 kΩ ±5 percent.
3. $C_{CL}$ includes scope capacitance.

FIGURE 2. Load test circuit.

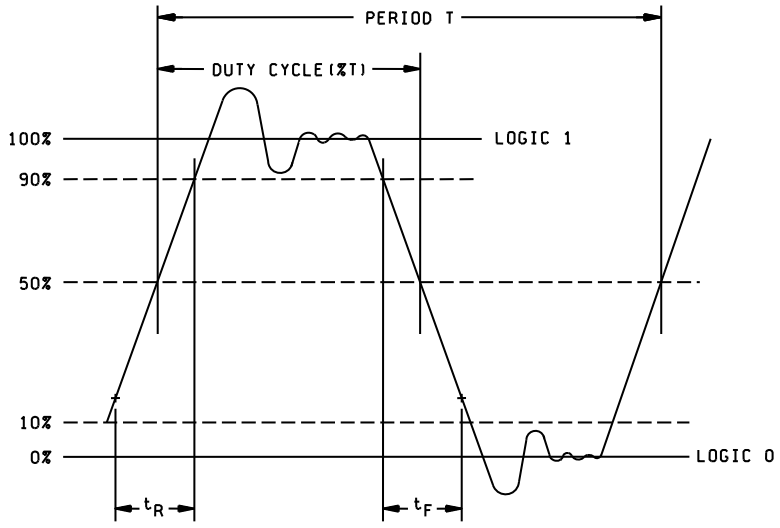



FIGURE 3. Waveform



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Frequency aging: Measurements shall be taken at +70°C ±0.2°C at intervals of not more than every 72 hours for 30 days minimum (Except one maximum interval of 96 hours per 30 days is permitted).

±10 ppm per year, maximum
±1.5 ppm per 30 days.
±3 ppm per 90 days.

Terminal strength: Method 211 of MIL-STD-202, test condition A.

Applied force: 8 ounces each terminal.

Frequency-environmental tolerance: Not applicable.

Vibration, sinusoidal: In accordance with MIL-PRF-55310 and method 204 of MIL-STD-202.

Nonoperating: Test condition G.

Operating: Not required.

Ambient pressure:

Nonoperating: In accordance with MIL-PRF-55310.

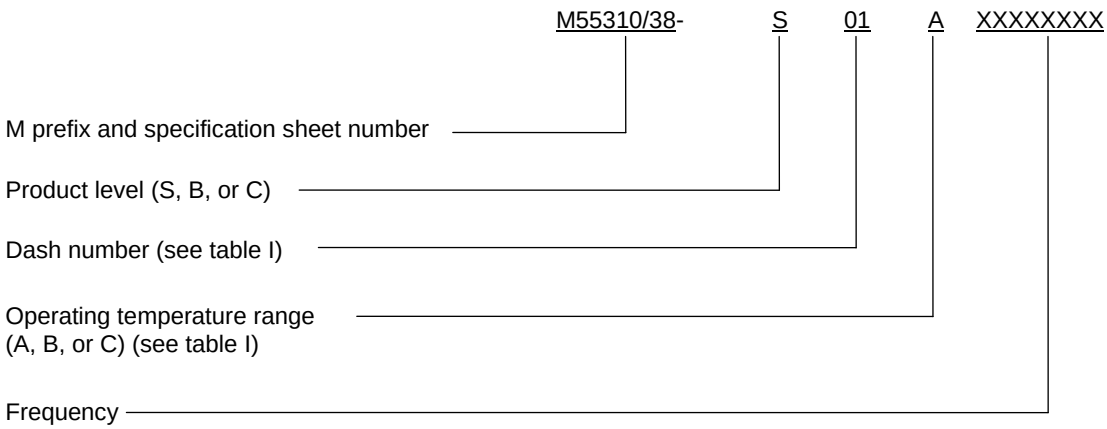
Operating: Method 105 of MIL-STD-202, test condition C.

Exposure time: 5 minutes.

Reflow soldering: Reflow soldering of the unit at +230°C ±10°C for 15 seconds shall not degrade the performance.

Part or Identifying Number (PIN): Consists of "M" prefix followed by specification sheet number, a dash and coded alphas, and numeric number. See example:

EXAMPLE




Referenced documents. In addition to MIL-PRF-55310, this document references MIL-STD-202.



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Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19
NASA - NA

Preparing activity:
Army - CR

Agent:
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